

Credit Card Fraud Detection — Assignment Answers

Question 1 | Five Data Points Required for Credit Card Fraud Detection (3 marks)

The following five data points are essential for detecting credit card fraud:

- **IP Address:** Tracks the origin of each transaction. Sudden changes in IP address — especially across different countries or regions — can indicate that a fraudster has taken over an account. Comparing IPs across sessions helps flag account takeovers.
- **Transaction Date and Time:** The timing of purchases is critical. Fraudulent transactions often occur at unusual hours (e.g., late at night or in the early morning) and may cluster within a very short time window, suggesting automated or rapid fraudulent activity.
- **Transaction Value:** Legitimate purchasing behaviour typically shows consistent, moderate spending patterns. Sudden large spikes in transaction value — especially across multiple purchases in quick succession — are a strong indicator of fraud or stolen card use.
- **Shipping Address:** Discrepancies between the account's registered address and the shipping address (such as a P.O. Box or an entirely different location) can signal fraud. Fraudsters frequently redirect goods to untraceable addresses.
- **Units Purchased:** Unusually large quantities purchased in a single transaction, particularly for high-value goods like electronics or tools, is a common fraud pattern, often used for resale. Comparing order quantities to historical averages helps detect anomalies.

Question 2 | Three Errors/Issues Impacting Data Accuracy (3 marks)

The following three data quality issues were identified in the table:

- **Missing IP Address (johnp — Row 5):** One of johnp's transactions dated 2020-06-03 at 01:15:12 has a completely blank IP address field. This is a data completeness error. Without an IP address, it is impossible to trace the origin of the transaction or determine whether it originated from a legitimate or malicious source, reducing the reliability of any fraud analysis.
- **Missing Transaction Value (johnp — Row 3):** The transaction for johnp dated 1-6-20 in the Home Décor category has no transaction value recorded. This is a data completeness/accuracy issue. Missing monetary values prevent accurate

calculation of total spend, skew statistical analyses, and make it impossible to assess whether the transaction amount was normal or anomalous.

- **Inconsistent Date Formatting:** Transaction dates appear in multiple formats across the table — for example: '15-5-20', '10-6-20', '2020-06-03', '15 May 2020', and '07 June 2020'. This inconsistency is a data standardisation error. Mixed formats complicate automated sorting, filtering, and time-series analysis, and can cause errors when data is processed by software systems that expect a single consistent format.

Question 3 | Two Anomalies Suggesting Suspicious Transactions (2 marks)

Two notable anomalies were identified from the transaction table:

- **Anomaly 1 — ellend's Sudden High-Value Laptop Purchase:** User ellend made two small, routine purchases: \$99.99 (Clothing) on 07 June 2020 and \$53.15 (Beauty) on 08 June 2020. Then, on 02 July 2020, a transaction of \$4,895.00 for a Laptop appears — more than 40 times the value of previous purchases. This dramatic value spike is highly anomalous. Furthermore, this transaction originated from a different IP address (1.167.255.10 vs. the usual 172.165.10.1) and was shipped to a different address (P.O. Box 5401 vs. the usual P.O. Box 1322). The combination of an IP change, address change, and extreme value spike strongly suggests possible account takeover or fraudulent use.
- **Anomaly 2 — davidg's Missing IP Address on a Purchase:** User davidg has a transaction on 01 June 2020 for \$157.25 (Car Spares, 2 units) with a completely blank IP address field. All other davidg transactions show the IP address 1.58.167.2. The sudden absence of an IP address on a specific transaction is unusual and may indicate the transaction was processed through an anonymised or masked network connection, which is a known fraud evasion tactic.

Question 4 | Key Takeaway from the Transaction Value Chart (1 mark)

The line chart 'Transaction Value per Transaction' reveals a striking contrast between the spending patterns of johnp and davidg compared to ellend.

Johnp's transaction values rise sharply from Trn#1 through Trn#3, reaching a peak at Trn#3 with an approximate value of \$4,131, before slightly declining at Trn#4 and Trn#5. This steep escalation across five consecutive transactions is a strong fraud indicator, as it is inconsistent with normal consumer purchasing behaviour.

Davidg's transactions remain relatively low and flat, showing modest and consistent spending across all five transactions — a pattern more typical of legitimate purchasing behaviour.

Overall, the chart makes it visually clear that johnp's accelerating transaction values stand out as anomalous, reinforcing the conclusion that their account activity is suspicious and potentially fraudulent.

Question 5 | Type of Analytics Used for Fraud Detection (1 mark)

When analysing historical credit card data to understand what a fraudulent transaction looks like, the type of analytics being performed is **Descriptive Analytics**.

Descriptive analytics focuses on examining and summarising historical data to understand patterns and trends that have already occurred. In the context of credit card fraud, this means reviewing past transaction records — such as IP addresses, transaction amounts, purchase frequency, and shipping addresses — to identify what characteristics are associated with fraudulent behaviour.

This is distinct from the other types of analytics:

- **Diagnostic Analytics:** asks why something happened (e.g., why did fraud spike in June?).
- **Predictive Analytics:** uses historical patterns to forecast future fraud events.
- **Prescriptive Analytics:** recommends actions to prevent or reduce fraud going forward.

Since the task involves reviewing what has already happened in the data to recognise fraud patterns, Descriptive Analytics is the correct classification.